

Tense Timing Experiments: Analysis & Report

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Report

Three experiments test whether tense is planned before articulation begins, and whether the cost of that planning is morphosyntactic (selecting a tense feature) or morphophonological (retrieving the form that realises it). Experiment 1 uses a speeded keypress decision; Experiments 2 and 3 use spoken production, with timing recovered from forced-aligned recordings.

All timing measures are in milliseconds. Error bars are 95% intervals built on Morey (2008) within-subject standard errors. Conditions are distinguished by marker fill and line type rather than colour, so the figures survive greyscale printing.

Data and preprocessing

Spoken responses were uploaded per block as .webm, converted to 16 kHz mono wav, transcribed with AssemblyAI, and force-aligned against those transcripts with the Montreal Forced Aligner (english_us_arpa acoustic model and dictionary, out-of-vocabulary items passed through G2P). Word and phone boundaries come from the resulting TextGrids. The commands are listed in the appendix.

Recordings that never reached the server, and sessions with no usable audio, are outside the participant screen below: they simply have no timing data. One morphophonology participant uploaded 101 empty five-byte files and contributes no recordings at all.

Participant filtering

Forced alignment yielded 48 morphophonology participants and 24 morphosyntax participants. Exclusions were applied in the order below; each step acts only on the participants that survived the preceding steps.

Step 1, low accuracy. Participants who passed fewer than 50% of trials on the utterance criteria were excluded: 7 of 48 in morphophonology (41 remaining), and 7 of 24 in morphosyntax (17 remaining).

After all participant-level exclusions, 41 morphophonology participants and 17 morphosyntax participants remain.

Within the retained participants, individual trials were excluded when the response did not name the correct subject entity, did not use the target verb in the tense the trial called for, did not match the target sentence in length, was empty, or marked tense with an adverbial instead of on the verb. This retained 3799 of 4344 trials (87%) in morphophonology and 1400 of 1800 trials (78%) in morphosyntax.

Experiment 1: Conceptual task

Prediction

Repetition should have no effect here. If the repetition effects in the production experiments come from reusing linguistic material, then a task that requires no verb form to be produced gives nothing to reuse. Participants decide which time a picture refers to and press a key. Deciding about the future may take longer than deciding about the past, but repeating a decision about the same time should not speed up the next one, and the effect of repetition should not differ between the two tenses. A repetition effect here would mean the effects in the other two experiments need not be linguistic at all.

106 participants learned character–verb–time pairings (Chef, Pirate, Wizard x 20 verbs) and made speeded keypress decisions (C = Past, M = Future). The manipulation is **tense priming**: whether the current trial's tense matches the previous trial's (primed) or switches (unprimed). No speech is recorded, so the utterance screen does not apply here.

Filtering

Of 11880 trials, 3960 boundary trials (positions 1, 6, 7 and 12 of each 12-trial block) were excluded. A further 493 trials fell outside the 200–5000 ms response window and were dropped as anticipations or disengagement: the raw responses run from 0 ms to 173 s, and the upper tail distorts both the means and the lognormal fit. That leaves **7427 trials**.

Means

Table 1: Experiment 1: decision RT by tense x prime status

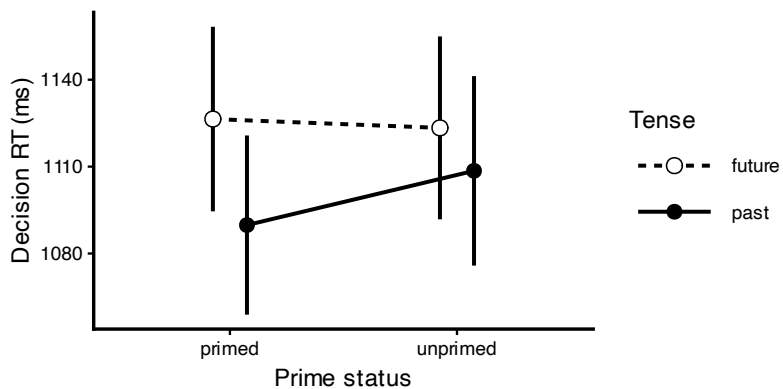
Tense	Prime status	N	Mean RT (ms)	SE
future	primed	1855	1126	16.1
future	unprimed	1860	1123	16.1
past	primed	1859	1090	15.6
past	unprimed	1853	1109	16.7

Regularity repetition is also tracked. The two are not independent: regularity repeats on 50% of trials where tense repeats but 42% of trials where it switches, so the two priming effects are estimated jointly in one model rather than separately.

Table 2: Experiment 1: decision RT by regularity prime x tense prime

Regularity prime	Tense prime	N	Mean RT (ms)	SE
primed	primed	1860	1091	15.1
primed	unprimed	1539	1183	19.2
unprimed	primed	1854	1125	16.6
unprimed	unprimed	2174	1069	14.3

Plot



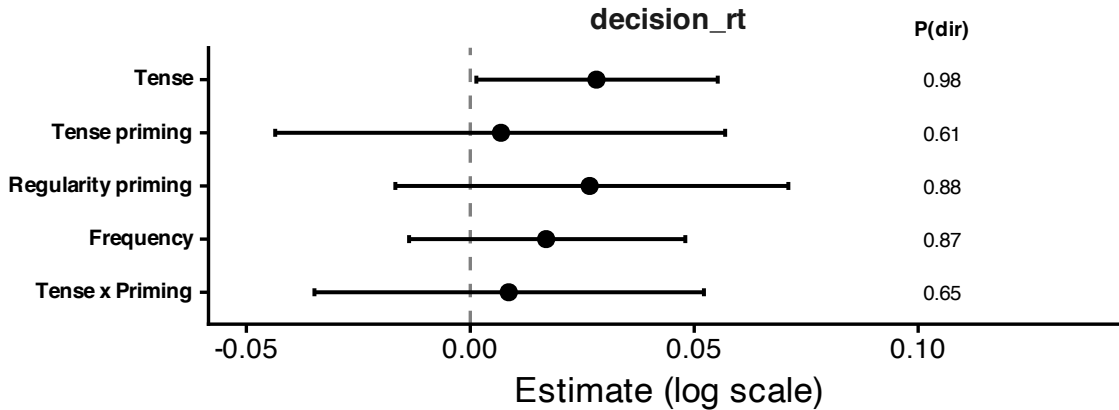
Modelling

The model estimates each measure separately. Positive coefficients mean a later onset or a longer duration. The measures are the time from the start of the trial to the first word, the time to the subject noun, the duration of the subject noun per phone, the duration of the verb per phone, and the interval between the subject noun and the verb.

Speakers reach the verb without pausing on most trials, so the interval is modelled in two parts. **Gap length** is how long the pause lasts on the trials where there is one. **No pause lik.** is how likely the speaker is to run straight into the verb; a positive coefficient there means fewer pauses. The first part is on the log scale in milliseconds, the second on the log-odds scale.

Table 3: Experiment 1, reported model (spec 5). Positive = slower/longer.

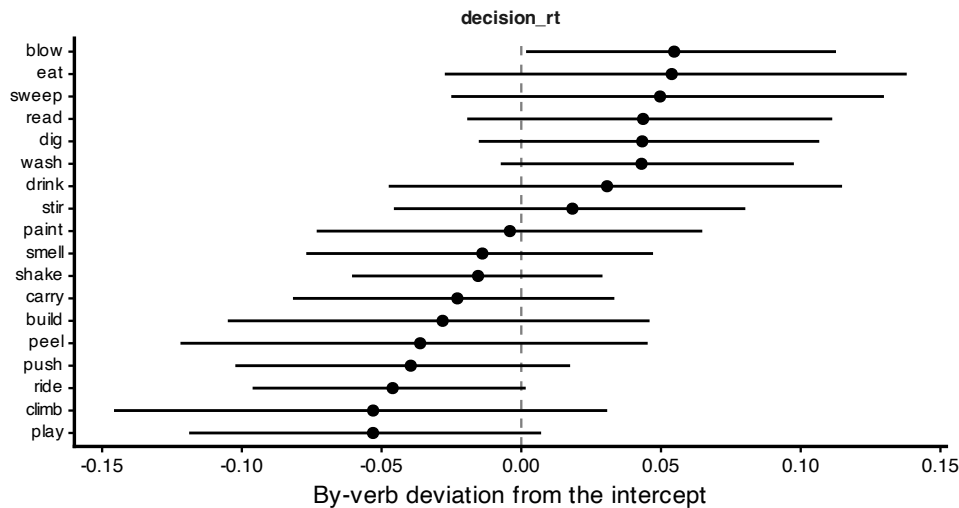
Term	Estimate	95% CrI	P(dir)
decision_rt			
Tense priming	0.007	[-0.044, 0.057]	0.61
Regularity priming	0.027	[-0.017, 0.071]	0.88
Tense	0.028	[0.001, 0.055]	0.98
Tense x Priming	0.009	[-0.035, 0.052]	0.65
Pirate vs Chef	-0.045	[-0.072, -0.018]	> 0.99
Wizard vs Chef	0.012	[-0.014, 0.040]	0.82
Frequency	0.017	[-0.014, 0.048]	0.87



Items

By-verb deviations from the intercept in the reported model. A verb whose interval excludes zero behaves differently from the rest of the item set.

Experiment 1



Results

Decisions were slower for future than for past. Future trials averaged 1125 ms (SE 11.4) and past trials 1099 ms (SE 11.4). The model supports this difference ($\beta = 0.028$, 95% CrI [0.001, 0.055], $P_{dir} = 0.98$).

Tense repetition did not affect decision times. Primed trials averaged 1108 ms (SE 11.2) and unprimed trials 1116 ms (SE 11.6), and the model gives $\beta = 0.007$, 95% CrI [-0.044, 0.057], $P_{dir} = 0.61$. The interaction between tense and tense repetition is absent ($\beta = 0.009$, 95% CrI [-0.035, 0.052], $P_{dir} = 0.65$), with the posterior centred on zero.

This is what we expected. The conceptual task requires participants to retrieve which time a picture refers to and press a key, with no verb form to produce. Deciding about the future takes longer than deciding about the past, but repeating a decision about the same time does not make the next one faster.

Experiment 2: Morphophonology

Prediction

Repetition should act near the verb. The form of the past tense is not needed until the verb is articulated, so preparing it should leave the time to begin speaking unchanged and appear on the words immediately around the verb. The effect should be larger for irregular verbs, whose past forms have to be retrieved individually, than for regular verbs, which share one ending. The clearest place to see this is the interaction between regularity and repetition, on the duration of the subject noun and on whether the speaker pauses before reaching the verb.

41 participants produced past-tense descriptions of the same scenes (*The pirate spun a top*). The manipulation is **morphophonological priming** by verb regularity: whether the current trial's verb shares its past formation with the previous trial's. All trials are past tense, so tense is held constant and regularity carries the design.

Measures. Utterance onset (when *the* begins), NP onset, NP head duration (the entity name), and the gap between NP offset and the verb.

Filtering

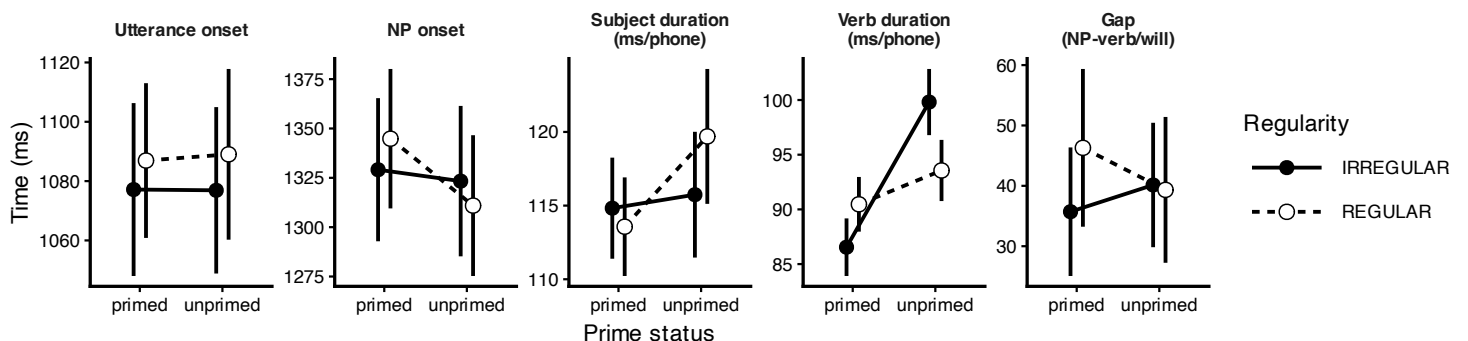
Of 4344 trials, 545 (13%) failed the utterance screen and 1448 were boundary trials, leaving **2535 trials** from 41 participants.

Means

Table 4: Experiment 2: timing measures by regularity x prime status

Measure	Condition	Prime	N	Mean	SE	Measure	Condition	Prime	N	Mean	SE
Utterance onset	IRREGULAR	primed	612	1077	14.9	Subject duration (ms/phone)	IRREGULAR	primed	630	115	1.8
	IRREGULAR	unprimed	635	1077	14.3		IRREGULAR	unprimed	649	116	2.2
	REGULAR	primed	622	1087	13.3		REGULAR	primed	642	114	1.7
	REGULAR	unprimed	598	1089	14.7		REGULAR	unprimed	614	120	2.3
NP onset	IRREGULAR	primed	630	1329	18.5	Verb duration (ms/phone)	IRREGULAR	primed	550	87	1.3
	IRREGULAR	unprimed	649	1323	19.4		IRREGULAR	unprimed	649	100	1.5
	REGULAR	primed	642	1345	18.0		REGULAR	primed	642	90	1.3
	REGULAR	unprimed	614	1311	18.2		REGULAR	unprimed	602	94	1.4
Gap (NP-verb/will)							IRREGULAR	primed	630	36	5.4
							IRREGULAR	unprimed	649	40	5.3
							REGULAR	primed	642	46	6.7
							REGULAR	unprimed	614	39	6.2

Plot



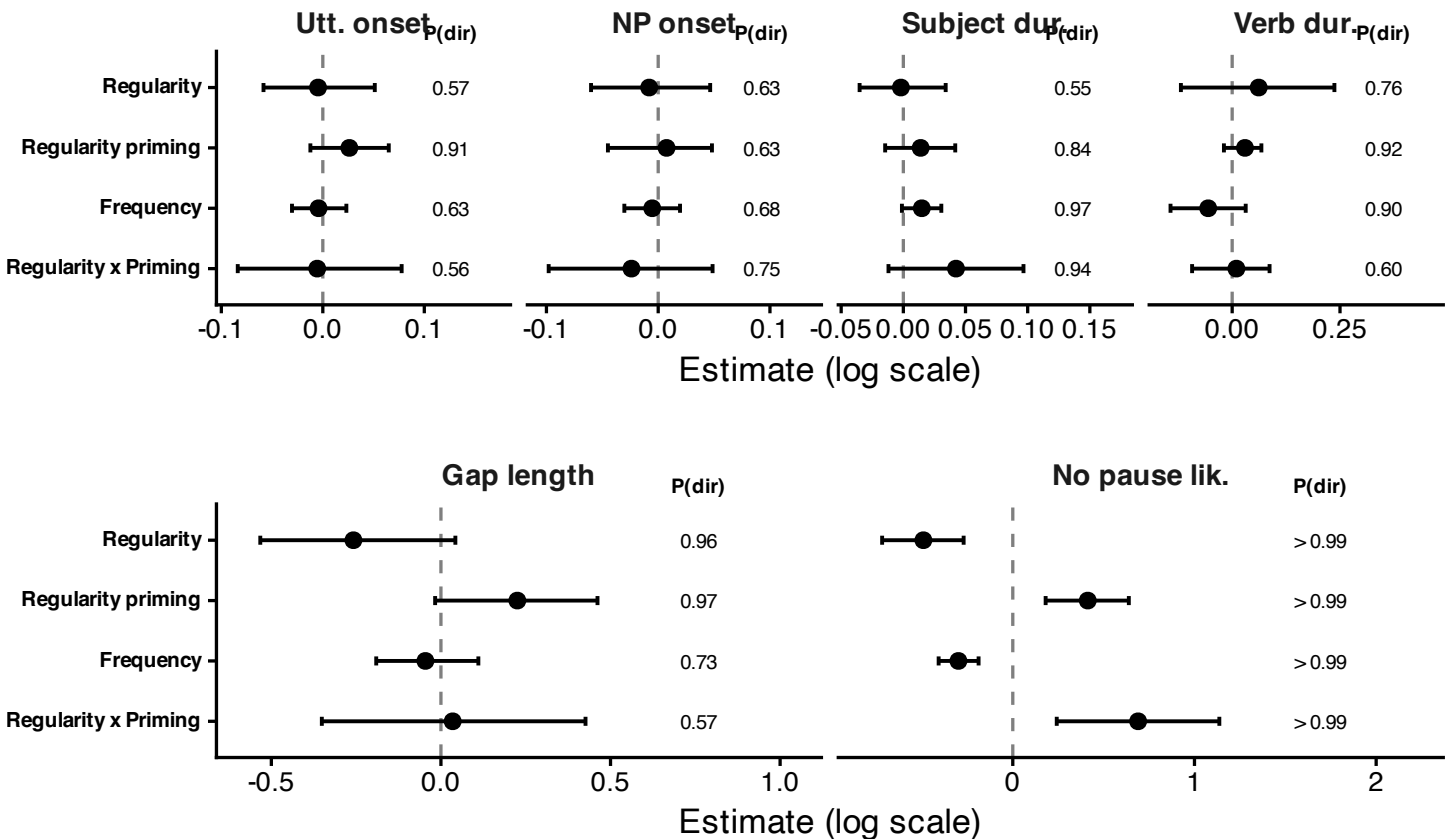
Modelling

The model estimates each measure separately. Positive coefficients mean a later onset or a longer duration. The measures are the time from the start of the trial to the first word, the time to the subject noun, the duration of the subject noun per phone, the duration of the verb per phone, and the interval between the subject noun and the verb.

Speakers reach the verb without pausing on most trials, so the interval is modelled in two parts. **Gap length** is how long the pause lasts on the trials where there is one. **No pause lik.** is how likely the speaker is to run straight into the verb; a positive coefficient there means fewer pauses. The first part is on the log scale in milliseconds, the second on the log-odds scale.

Table 5: Experiment 2, reported model (spec 5). Positive = slower/longer.

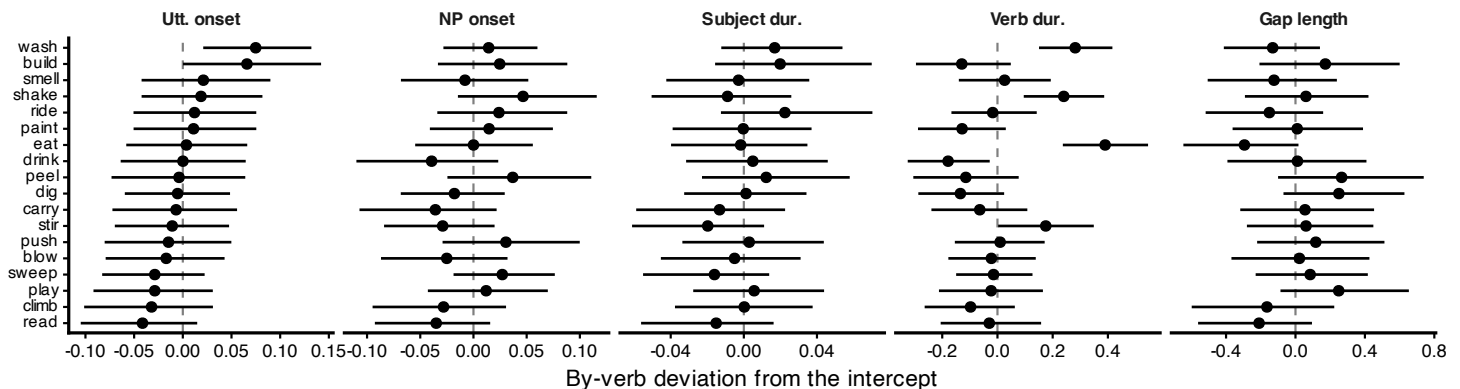
Measure	Term	Est.	95% CrI	P(dir)	Measure	Term	Est.	95% CrI	P(dir)
Utt. onset	Regularity priming	0.026	[-0.012, 0.065]	0.91*	Verb dur.	Regularity priming	0.030	[-0.019, 0.068]	0.92*
	Regularity	-0.005	[-0.058, 0.051]	0.57		Regularity	0.061	[-0.119, 0.237]	0.76
	Regularity x Priming	-0.006	[-0.084, 0.078]	0.56		Regularity x Priming	0.010	[-0.093, 0.087]	0.60
	Pirate vs Chef	0.036	[0.003, 0.068]	0.99*		Pirate vs Chef	-0.024	[-0.051, 0.004]	0.96*
	Wizard vs Chef	-0.029	[-0.060, 0.002]	0.97*		Wizard vs Chef	-0.011	[-0.037, 0.015]	0.80
	Frequency	-0.004	[-0.030, 0.023]	0.63		Frequency	-0.056	[-0.143, 0.031]	0.90*
NP onset	Articulation rate	0.094	[0.066, 0.121]	> 0.99*	Gap length	Articulation rate	-0.009	[-0.033, 0.014]	0.78
	Regularity priming	0.007	[-0.045, 0.048]	0.63		Regularity priming	0.225	[-0.017, 0.462]	0.97*
	Regularity	-0.008	[-0.060, 0.047]	0.63		Regularity	-0.258	[-0.533, 0.043]	0.96*
	Regularity x Priming	-0.024	[-0.098, 0.049]	0.75		Regularity x Priming	0.035	[-0.351, 0.426]	0.57
	Pirate vs Chef	0.064	[0.038, 0.092]	> 0.99*		Pirate vs Chef	-0.168	[-0.353, 0.017]	0.96*
	Wizard vs Chef	-0.015	[-0.040, 0.009]	0.89		Wizard vs Chef	-0.295	[-0.479, -0.109]	> 0.99*
Subject dur.	Frequency	-0.005	[-0.030, 0.020]	0.68	No pause lik.	Frequency	-0.046	[-0.191, 0.110]	0.73
	Articulation rate	-0.121	[-0.143, -0.099]	> 0.99*		Articulation rate	0.111	[-0.071, 0.288]	0.89
	Regularity priming	0.014	[-0.015, 0.042]	0.84		Regularity priming	0.413	[0.181, 0.639]	> 0.99*
	Regularity	-0.002	[-0.035, 0.034]	0.55		Regularity	-0.492	[-0.719, -0.271]	> 0.99*
	Regularity x Priming	0.042	[-0.012, 0.097]	0.94*		Regularity x Priming	0.690	[0.242, 1.137]	> 0.99*
	Pirate vs Chef	-0.319	[-0.345, -0.293]	> 0.99*		Pirate vs Chef	0.305	[0.045, 0.575]	0.99*
	Wizard vs Chef	-0.406	[-0.431, -0.381]	> 0.99*		Wizard vs Chef	0.297	[0.042, 0.556]	0.99*
	Frequency	0.015	[-0.001, 0.031]	0.97*		Frequency	-0.299	[-0.408, -0.188]	> 0.99*
	Articulation rate	0.012	[-0.009, 0.035]	0.87		Articulation rate	0.323	[0.098, 0.548]	> 0.99*



Items

By-verb deviations from the intercept in the reported model. A verb whose interval excludes zero behaves differently from the rest of the item set.

Experiment 2



Results

Repetition of the past-tense form shortened the verb. After an irregular prime, irregular verbs were produced in 87 ms/phone (SE 1.5) against 100 ms/phone (SE 1.7) when unprimed. Regular verbs changed less, from 94 ms/phone (SE 1.4) to 90 ms/phone (SE 1.3). The model supports the repetition effect ($\beta = 0.030$, 95% CrI [-0.019, 0.068], $P_{dir} = 0.92$) but does not separate the two verb types ($\beta = 0.010$, 95% CrI [-0.093, 0.087], $P_{dir} = 0.60$).

Repetition also changed the subject noun, and here the two verb types behave differently. Before a regular verb, the subject was 120 ms/phone (SE 2.1) when unprimed and 114 ms/phone (SE 1.6) when primed. Before an irregular verb the two were close, 116 ms/phone (SE 2.0) and 115 ms/phone (SE 1.7). The model gives $\beta = 0.042$, 95% CrI [-0.012, 0.097], $P_{dir} = 0.94$ for the interaction, which we read as support given the direction and the size of the posterior mass above zero.

The clearest effect is on whether speakers paused at all before the verb. Pauses were most frequent before an unprimed irregular verb, on 28.2% (SE 1.8) of trials, and dropped to 18.6% (SE 1.5) when the preceding trial had also used an irregular. Regular verbs showed no such change, 16.1% (SE 1.5) when unprimed and 17.6% (SE 1.5) when primed. The model separates all three terms: irregular verbs are followed by more pauses ($\beta = -0.492$, 95% CrI [-0.719, -0.271], $P_{dir} = > 0.99$), repetition reduces pausing ($\beta = 0.413$, 95% CrI [0.181, 0.639], $P_{dir} = > 0.99$), and the reduction is larger for irregulars ($\beta = 0.690$, 95% CrI [0.242, 1.137], $P_{dir} = > 0.99$). The coefficients are on the probability of producing no pause, so a positive value means fewer pauses.

Speech onset was unaffected. Primed trials began at 1082 ms (SE 10.3) and unprimed trials at 1083 ms (SE 10.6) ($\beta = 0.026$, 95% CrI [-0.012, 0.065], $P_{dir} = 0.91$).

Repetition of the past-tense form therefore acts late. It leaves the time to begin speaking unchanged and shows up on the subject noun, on the verb itself, and on whether the speaker pauses before reaching the verb. This is what we predicted for a form-based effect: the form is not needed until the verb is articulated, so preparing it does not delay the start of the sentence.

Experiment 3: Morphosyntax

Prediction

Repetition should act early. Choosing between past and future is a choice about the structure of the sentence, made before articulation begins, so its effects should appear at the start of the utterance rather than around the verb. The interaction between tense and repetition at the onset of the subject noun is the measure this predicts, and the later measures should show little.

17 participants produced past or future descriptions of the same scenes (*The pirate spun a top* vs *The pirate will spin a top*). The manipulation is **tense priming**, identical in logic to Experiment 1, but the response is spoken, so the dependent measures are timing rather than keypress latency.

Measures. As in Experiment 2, with the gap running from NP offset to the tense-bearing word (*will* in the future, the past-marked verb in the past).

Filtering

Of 1800 trials, 400 (22%) failed the utterance screen and 600 were boundary trials, leaving **946 trials** from 17 participants.

Means

Table 6: Experiment 3: timing measures by tense x prime status

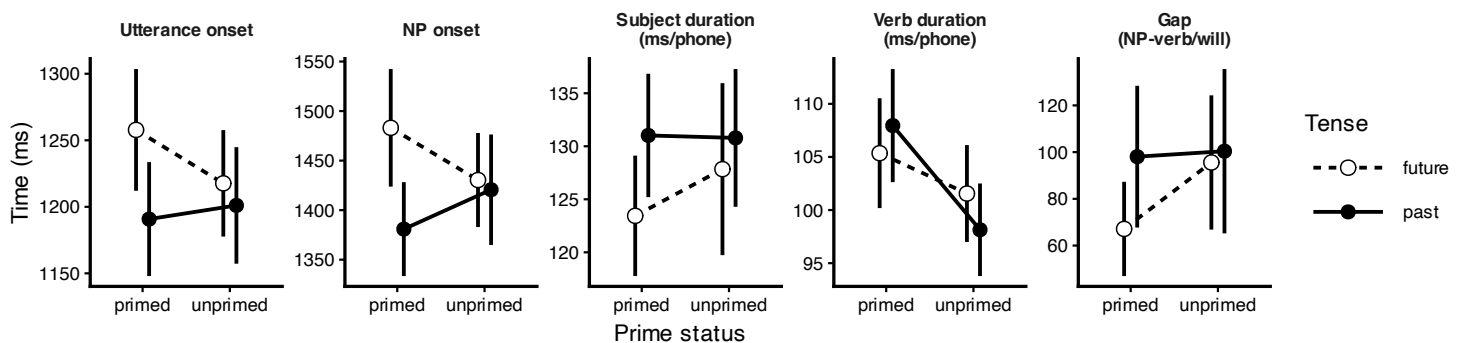
Measure	Condition	Prime	N	Mean	SE	Measure	Condition	Prime	N	Mean	SE
Utterance onset	future	primed	237	1258	23.3	Subject duration (ms/phone)	future	primed	240	123	2.9
	future	unprimed	246	1218	20.4		future	unprimed	246	128	4.1
	past	primed	230	1191	21.8		past	primed	231	131	3.0
	past	unprimed	229	1201	22.3		past	unprimed	229	131	3.3
NP onset	future	primed	240	1483	30.3	Verb duration (ms/phone)	future	primed	240	105	2.6
	future	unprimed	246	1430	24.2		future	unprimed	246	102	2.3
	past	primed	231	1381	24.2		past	primed	221	108	2.7
	past	unprimed	229	1421	28.4		past	unprimed	221	98	2.2
Gap (NP-verb/will)							future	primed	240	67	10.3
							future	unprimed	246	96	14.7
							past	primed	231	98	15.5
							past	unprimed	229	100	17.9

As in Experiment 1, regularity repetition is tracked alongside tense repetition and the two enter the models together.

Table 7: Experiment 3: timing by regularity prime x tense prime

Regularity prime	Tense prime	N	Utterance onset (ms)	NP head dur (ms)	Gap (ms)
primed	primed	237	1254	524	73
primed	unprimed	201	1230	550	93
unprimed	primed	234	1195	534	92
unprimed	unprimed	274	1195	517	101

Plot



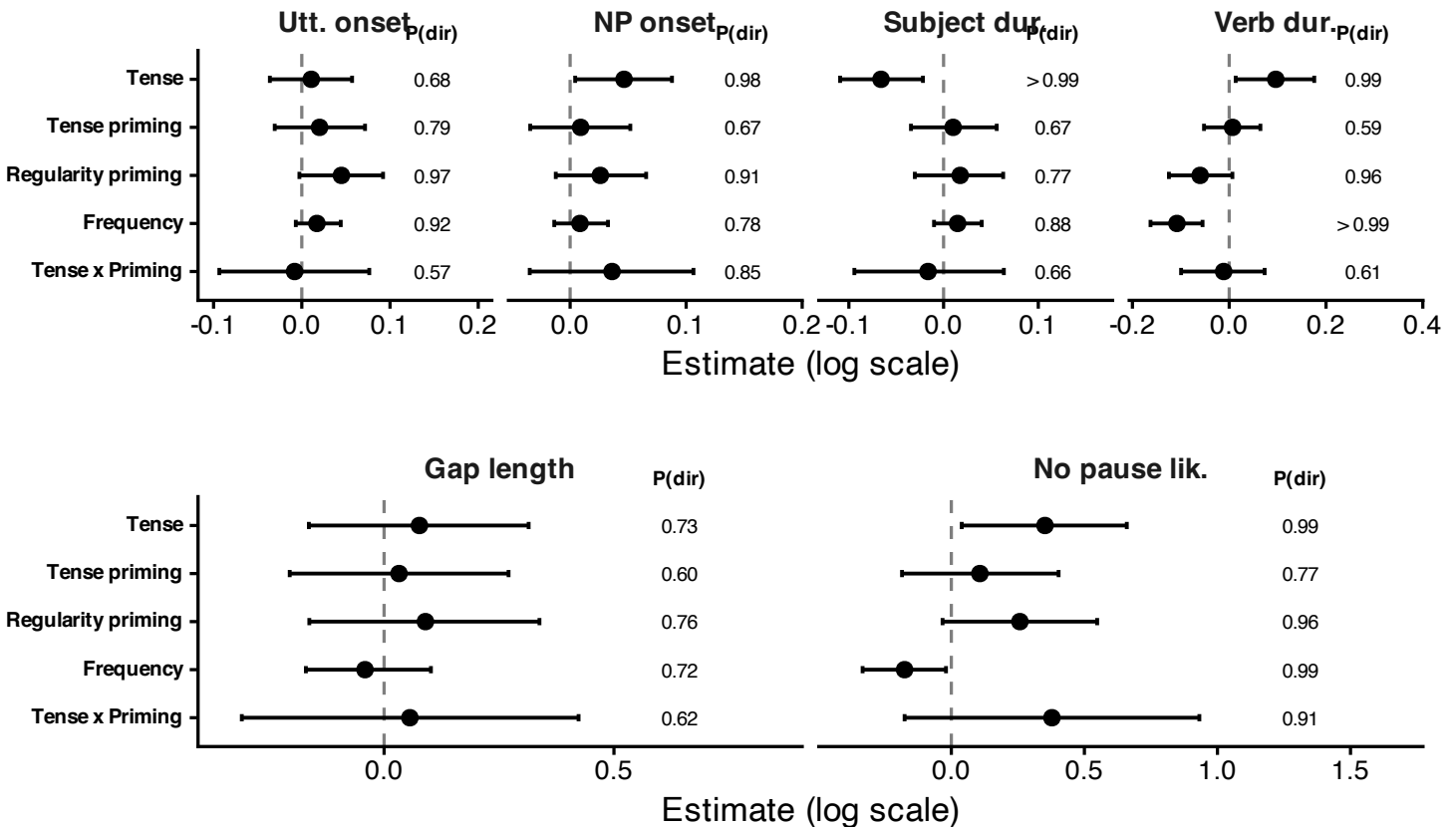
Modelling

The model estimates each measure separately. Positive coefficients mean a later onset or a longer duration. The measures are the time from the start of the trial to the first word, the time to the subject noun, the duration of the subject noun per phone, the duration of the verb per phone, and the interval between the subject noun and the verb.

Speakers reach the verb without pausing on most trials, so the interval is modelled in two parts. **Gap length** is how long the pause lasts on the trials where there is one. **No pause lik.** is how likely the speaker is to run straight into the verb; a positive coefficient there means fewer pauses. The first part is on the log scale in milliseconds, the second on the log-odds scale.

Table 8: Experiment 3, reported model (spec 5). Positive = slower/longer.

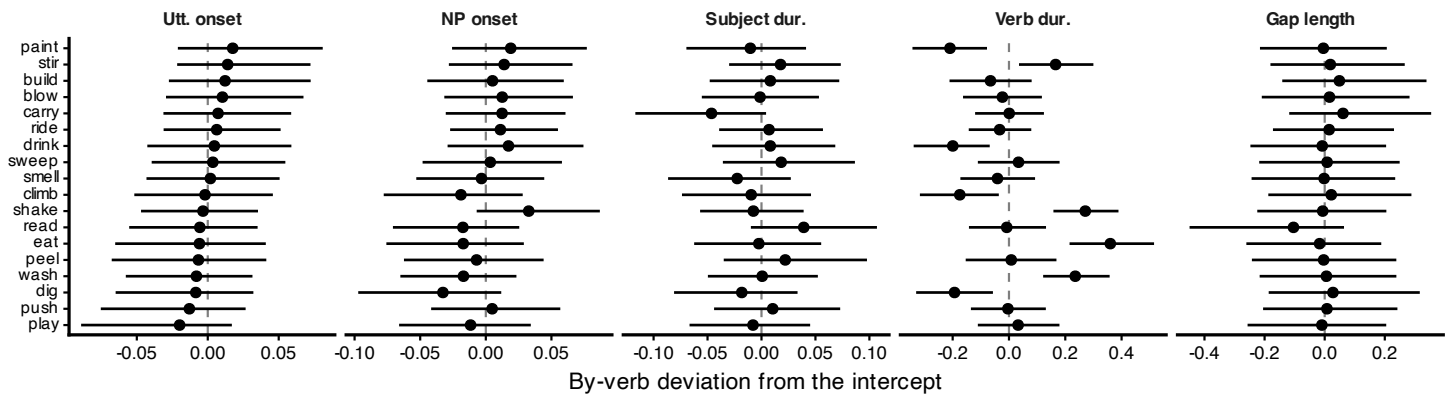
Measure	Term	Est.	95% CrI	P(dir)	Measure	Term	Est.	95% CrI	P(dir)
Utt. onset	Tense priming	0.020	[-0.031, 0.072]	0.79	Verb dur.	Tense priming	0.007	[-0.052, 0.064]	0.59
	Regularity priming	0.045	[-0.003, 0.092]	0.97*		Regularity priming	-0.060	[-0.125, 0.007]	0.96*
	Tense	0.011	[-0.036, 0.057]	0.68		Tense	0.096	[0.013, 0.176]	0.99*
	Tense x Priming	-0.008	[-0.093, 0.076]	0.57		Tense x Priming	-0.011	[-0.099, 0.073]	0.61
	Pirate vs Chef	-0.015	[-0.060, 0.030]	0.74		Pirate vs Chef	-0.015	[-0.064, 0.033]	0.73
	Wizard vs Chef	-0.053	[-0.098, -0.008]	0.99*		Wizard vs Chef	0.042	[-0.007, 0.090]	0.95*
	Frequency	0.017	[-0.007, 0.044]	0.92*		Frequency	-0.108	[-0.163, -0.055]	> 0.99*
NP onset	Articulation rate	0.026	[0.003, 0.049]	0.99*	Gap length	Articulation rate	-0.027	[-0.053, -0.001]	0.98*
	Tense priming	0.009	[-0.034, 0.052]	0.67		Tense priming	0.033	[-0.205, 0.271]	0.60
	Regularity priming	0.026	[-0.012, 0.066]	0.91*		Regularity priming	0.090	[-0.163, 0.338]	0.76
	Tense	0.046	[0.004, 0.088]	0.98*		Tense	0.077	[-0.164, 0.314]	0.73
	Tense x Priming	0.036	[-0.035, 0.106]	0.85		Tense x Priming	0.056	[-0.309, 0.423]	0.62
	Pirate vs Chef	0.019	[-0.017, 0.055]	0.84		Pirate vs Chef	-0.231	[-0.481, 0.026]	0.96*
	Wizard vs Chef	-0.030	[-0.067, 0.007]	0.95*		Wizard vs Chef	-0.280	[-0.531, -0.026]	0.99*
Subject dur.	Frequency	0.009	[-0.014, 0.033]	0.78	No pause lik.	Frequency	-0.041	[-0.170, 0.102]	0.72
	Articulation rate	-0.070	[-0.089, -0.052]	> 0.99*		Articulation rate	0.124	[-0.001, 0.249]	0.97*
	Tense priming	0.010	[-0.034, 0.056]	0.67		Tense priming	0.107	[-0.185, 0.403]	0.77
	Regularity priming	0.018	[-0.030, 0.063]	0.77		Regularity priming	0.258	[-0.033, 0.548]	0.96*
	Tense	-0.066	[-0.109, -0.022]	> 0.99*		Tense	0.352	[0.039, 0.659]	0.99*
	Tense x Priming	-0.016	[-0.094, 0.064]	0.66		Tense x Priming	0.378	[-0.176, 0.932]	0.91*
	Pirate vs Chef	-0.339	[-0.381, -0.296]	> 0.99*		Pirate vs Chef	-0.199	[-0.543, 0.149]	0.87
	Wizard vs Chef	-0.445	[-0.489, -0.402]	> 0.99*		Wizard vs Chef	-0.223	[-0.572, 0.126]	0.89
	Frequency	0.015	[-0.010, 0.040]	0.88		Frequency	-0.176	[-0.333, -0.020]	0.99*
	Articulation rate	0.008	[-0.014, 0.029]	0.76		Articulation rate	0.117	[-0.055, 0.289]	0.91*



Items

By-verb deviations from the intercept in the reported model. A verb whose interval excludes zero behaves differently from the rest of the item set.

Experiment 3



Results

Tense affected the sentence throughout. Future sentences reached the subject noun at 1456 ms (SE 20.8) against 1401 ms (SE 20.1) for past ($\beta = 0.046$, 95% CrI [0.004, 0.088], $P_{dir} = 0.98$). The subject noun was shorter in the future, 126 ms/phone (SE 2.3) against 131 ms/phone (SE 2.1) ($\beta = -0.066$, 95% CrI [-0.109, -0.022], $P_{dir} = > 0.99$), and the verb was longer, 103 ms/phone (SE 1.7) against 103 ms/phone (SE 1.7) ($\beta = 0.096$, 95% CrI [0.013, 0.176], $P_{dir} = 0.99$).

Tense repetition on its own did nothing. Primed trials reached the subject noun at 1433 ms (SE 21.8) and unprimed trials at 1426 ms (SE 19.3) ($\beta = 0.009$, 95% CrI [-0.034, 0.052], $P_{dir} = 0.67$), and the same holds for the other measures.

Repetition interacted with tense at the subject noun. In the future, primed trials reached the noun at 1483 ms (SE 33.5) and unprimed trials at 1430 ms (SE 24.9). In the past the ordering reverses, 1381 ms (SE 27.1) when primed and 1421 ms (SE 29.8) when unprimed. The model gives $\beta = 0.036$, 95% CrI [-0.035, 0.106], $P_{dir} = 0.85$. The posterior leans in the predicted direction but the interval is wide, and with 17 participants we treat this as suggestive rather than established. A replication would need roughly 45 participants to resolve it.

The interaction appears early, before the verb has been reached, which is where we expected an effect of tense selection to appear. Later measures do not show it: neither the duration of the subject noun ($\beta = -0.016$, 95% CrI [-0.094, 0.064], $P_{dir} = 0.66$) nor the duration of the verb ($\beta = -0.011$, 95% CrI [-0.099, 0.073], $P_{dir} = 0.61$) separates the conditions.

Appendix: model comparison and checks

Model comparison

Models were compared by leave-one-out cross-validation. Differences are relative to the best model for that measure; a difference smaller than about twice its standard error is not meaningful.

Table 9: Experiment 1: model comparison

measure	model	elpd_diff	se_diff
decision_rt	5: + Entity, rate	0.0	0.0
decision_rt	1: Tense x Prime	-6.3	4.7
decision_rt	2: + Frequency	-6.4	4.7

measure	model	elpd_diff	se_diff
decision_rt	3: + Reg. prime	-7.3	4.4

Table 10: Experiment 2: model comparison

measure	model	elpd_diff	se_diff
Utt. onset	2: + Entity, rate	0.0	0.0
Utt. onset	1: Regularity x Prime + Freq	-26.8	16.6
NP onset	2: + Entity, rate	0.0	0.0
NP onset	1: Regularity x Prime + Freq	-77.8	16.0
Subject dur.	2: + Entity, rate	0.0	0.0
Subject dur.	1: Regularity x Prime + Freq	-472.4	38.6
Verb dur.	1: Regularity x Prime + Freq	0.0	0.0
Verb dur.	2: + Entity, rate	-2.2	2.2
Gap length	2: + Entity, rate	0.0	0.0
Gap length	1: Regularity x Prime + Freq	-7.2	5.0

Table 11: Experiment 3: model comparison

measure	model	elpd_diff	se_diff
Utt. onset	5: + Entity, rate	0.0	0.0
Utt. onset	1: Tense x Prime	-1.2	4.1
Utt. onset	2: + Frequency	-1.6	3.7
Utt. onset	3: + Reg. prime	-2.1	3.2
NP onset	5: + Entity, rate	0.0	0.0
NP onset	1: Tense x Prime	-26.9	8.8
NP onset	2: + Frequency	-27.1	8.7
NP onset	3: + Reg. prime	-28.6	8.4
Subject dur.	5: + Entity, rate	0.0	0.0
Subject dur.	1: Tense x Prime	-192.2	23.5
Subject dur.	2: + Frequency	-192.8	23.5
Subject dur.	3: + Reg. prime	-194.3	23.6
Verb dur.	5: + Entity, rate	0.0	0.0
Verb dur.	2: + Frequency	-2.9	4.5
Verb dur.	3: + Reg. prime	-3.1	3.9
Verb dur.	1: Tense x Prime	-7.4	5.5
Gap length	5: + Entity, rate	0.0	0.0
Gap length	2: + Frequency	-1.5	3.9
Gap length	3: + Reg. prime	-2.6	3.5
Gap length	1: Tense x Prime	-2.6	4.6

Convergence

No divergent transitions, $\hat{R}$ at or below 1.01, and effective sample sizes in the thousands for every model reported.

Table 12: Experiment 1: convergence

measure	divergences	max_rhat	min_ess_bulk	min_ess_tail	ok
decision_rt	0	1.003	1116	2435	TRUE

Table 13: Experiment 2: convergence

measure	divergences	max_rhat	min_ess_bulk	min_ess_tail	ok
Utt. onset	0	1.005	946	2114	TRUE
NP onset	0	1.002	1814	2215	TRUE
Subject dur.	0	1.005	1068	1839	TRUE
Verb dur.	2	1.005	1055	1693	FALSE
Gap length	0	1.002	1839	1699	TRUE

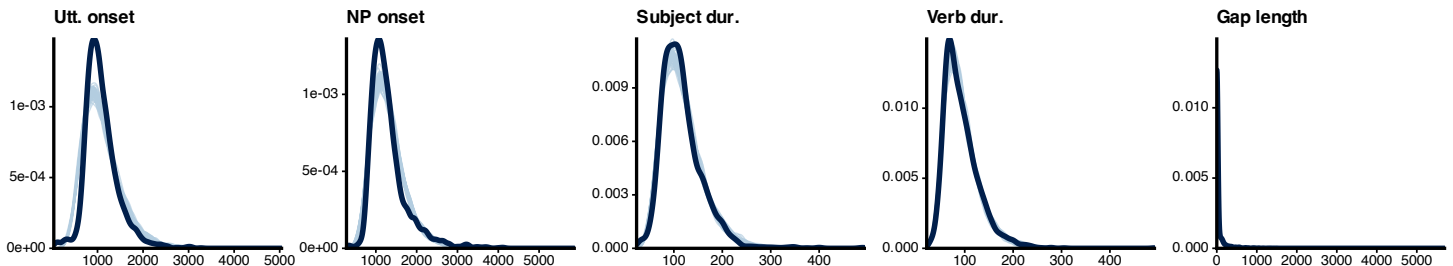
Table 14: Experiment 3: convergence

measure	divergences	max_rhat	min_ess_bulk	min_ess_tail	ok
Utt. onset	0	1.005	1573	3030	TRUE
NP onset	0	1.001	2048	3567	TRUE
Subject dur.	0	1.003	1729	2506	TRUE
Verb dur.	0	1.001	2906	5029	TRUE
Gap length	0	1.001	3548	4713	TRUE

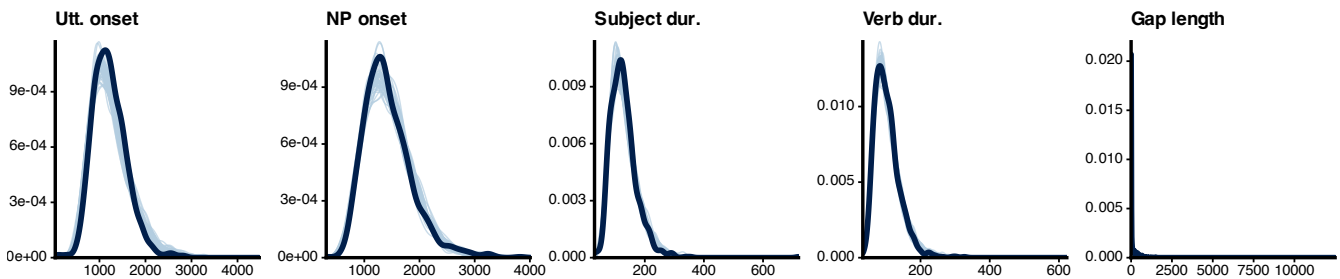
Posterior predictive checks

Each panel compares the observed distribution of a measure (dark line) against data simulated from the fitted model (light lines).

Experiment 2 - posterior predictive checks



Experiment 3 - posterior predictive checks



Experiment 1 - posterior predictive checks



Appendix: statistical models

All models were fitted with brms (Stan), four chains of 4000 iterations with the first 1000 discarded, `adapt_delta = 0.95`.

Contrast coding

All categorical predictors are sum coded at ± 0.5 , so each coefficient is the difference between the two levels and the intercept is the grand mean.

predictor	+0.5	-0.5	used in
Tense	future	past	Experiments 1 and 3
Regularity	irregular	regular	Experiment 2
Tense priming	primed	unprimed	Experiments 1 and 3
Regularity priming	primed	unprimed	all three

In Experiment 2 regularity priming is the manipulation and appears as *Priming* in the tables. In Experiments 1 and 3 the manipulation is tense priming, and regularity priming enters only as a control.

The subject noun (Chef, Pirate, Wizard) is treated as a factor with Chef as the reference level. Frequency is the rank of the produced word form in the SUBTLEX Zipf scale, from least to most frequent, centred and divided by its standard deviation. Ranking rather than the Zipf value itself keeps a small number of very frequent verbs from dominating an otherwise flat predictor. Articulation rate is the number of phones per second on that trial, computed over the words that are not being measured, and is also centred and scaled.

Response distributions

Onsets and durations are positive and right skewed and were fitted with a shifted lognormal distribution. The shift estimates a floor below which no response occurs.

The interval between the subject noun and the verb is zero on 80% of trials in Experiment 2 and 67% in Experiment 3, because speakers often move into the verb without pausing. A single continuous distribution describes this badly, so it was fitted with a hurdle lognormal: one part gives the probability of producing no pause, the other the duration of the pause when there is one. Both parts take the same predictors, which allows repetition to change how often speakers pause separately from how long they pause. Coefficients labelled as being on the probability of no pause come from the first part, where a positive value means fewer pauses.

Model structure

Every measure was fitted with the same structure. For Experiments 1 and 3,

$$\begin{aligned} \text{measure} \sim & \text{Tense} \times \text{Tense priming} + \text{Regularity priming} + \text{Frequency} + \text{Character} + \text{Rate} \\ & + (1 + \text{Tense} \times \text{Tense priming} + \text{Regularity priming} \mid \text{participant}) + (1 + \text{Tense priming} \mid \text{verb}) \end{aligned}$$

and for Experiment 2,

$$\begin{aligned} \text{measure} \sim & \text{Regularity} \times \text{Priming} + \text{Frequency} + \text{Character} + \text{Rate} \\ & + (1 + \text{Regularity} \times \text{Priming} \mid \text{participant}) + (1 + \text{Priming} \mid \text{verb}) \end{aligned}$$

Experiment 1 has no recordings and therefore no rate term.

Each measure was fitted with the same population-level terms: the design factor (tense or regularity), repetition, their interaction, frequency, the subject noun, articulation rate, and, in Experiments 1 and 3, form repetition as a control. Random effects are by participant, with slopes for the design factor, repetition and their interaction, and by verb, with a slope for

repetition. Frequency, the subject noun and articulation rate vary between items or within a trial and take no by-participant slope.

Regularity priming is included because it is not independent of tense priming in these materials. The form repeats on 50% of trials where tense repeats and 42% of trials where it switches, so leaving it out would let part of a form-repetition effect appear as a tense priming effect. It is never entered into an interaction.

Priors

parameter	prior
intercept	Normal(0, 1)
population-level coefficients	Normal(0, 0.3)
random-effect standard deviations	Normal(0, 0.3)
random-effect correlations	LKJ(2)
hurdle intercept and coefficients	Normal(0, 1)

Coefficients are on the log scale, so Normal(0, 0.3) places most of its mass on effects smaller than a factor of two and rules out nothing the data could show.

Reporting

Each coefficient is reported as its posterior median, a 95% credible interval, and P_{dir} , the proportion of the posterior lying on the same side of zero as the median. P_{dir} runs from 0.5, meaning the posterior is split evenly and the sign is undetermined, to 1, meaning the whole posterior falls on one side. It is not a significance test and has no threshold attached; we describe an effect as supported when most of the posterior is on one side and the interval is narrow enough to be informative. Values are never printed as 0 or 1, since a finite posterior sample cannot establish either.

Convergence and model checking

Every model reported here converged: no divergent transitions, $\hat{R} \leq 1.01$, and bulk effective sample sizes above 400. Posterior predictive checks compare simulated data against the observed distribution for each measure. Posterior contraction, the reduction in variance from prior to posterior, is above 0.94 for the frequency, subject noun and articulation rate terms and between 0.56 and 0.99 for the interactions, so the posteriors are determined by the data rather than by the priors.

Models were compared by leave-one-out cross-validation. Adding the subject noun and articulation rate improves prediction on the duration measures by a wide margin, which is why they are in the reported model: the three subject nouns differ by more than 50 ms per phone, far more than any effect of the manipulation.

Appendix: recordings and measurement

Spoken responses were recorded in the browser, one file per trial, and uploaded in blocks. Recordings were converted to 16 kHz mono audio and transcribed automatically. Each transcript was then aligned to its recording with the Montreal Forced Aligner using its English acoustic model and pronunciation dictionary, with pronunciations generated for words outside the dictionary. The alignment gives the start and end time of every word and every phone, and all timing measures are read from it.

Speech onset is the start of the first word. The onset of the subject noun and its duration are taken from the second word. The interval before the verb runs from the end of the subject noun to the start of the verb, and is zero when the two are adjacent. The verb is located by matching the transcript against the form the trial called for rather than by position, since the future adds a word before it.

Durations of the subject noun and the verb are divided by the number of phones the aligner placed inside them. Words differ in length, and without this a longer word would count as slower speech. Articulation rate is the number of phones per second over the remaining words of the sentence, which leaves the two measured words out of their own control variable.

Transcripts were scored against the sentence the trial called for. A response was kept if it named the right character and used the right verb in the right tense. Material after the verb was not scored, since every measure lies at or before the verb. Two adjustments were made after inspecting the transcripts. The past tense of *blow* is a homophone of *blue* and was transcribed as the colour on most trials, so both spellings were accepted. *Blowed* was kept as an error, being a real regularisation rather than a transcription problem. Responses that marked time with an adverbial instead of on the verb, such as *the wizard climbs a ladder in the past*, were excluded, since they leave nothing for the verb to carry.

Participants were excluded if fewer than half their responses met these criteria.